Thomas Richards

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Education

Northeastern University, Boston, MA

M.S. in Mechanical Engineering, General Mechanical Engineering, GPA: 3.91 — Sep. 2025 – April 2026

B.S. in Mechanical Engineering, Minor in Mathematics, GPA: 3.56 — Sep. 2021 – Aug. 2025

Courses: Control Systems Engineering, Mechatronic Systems, Wearable Robotics, Combustion & Air Pollution, Gas Turbine Combustion, Materials Process Selection, Thermal System Analysis & Design, Computation & Design

Activities: American Society of Mechanical Engineers (ASME), Northeastern Electric Racing (NER)

Technical Skills & Languages

Applications: SolidWorks (CSWP ID C-9HS7EXW6FK), AutoCAD, Fusion360, Agile PLM, SolidWorks PDM, MATLAB, Simulink, Simscape, Arduino, Ultimaker Cura, PrusaSlicer, Bambu Studio, Microsoft Office

Fabrication: 3D printing (FDM, SLA), milling, laser cutter, drill press, bandsaw, soldering, hand and power tools

Work Experience

Instron, Norwood, MA

Mechanical Engineering Co-op, R&D Team — January 2024 – June 2024

- Systematically diagnosed issues to identify the root cause of performance and functionality problems
- Conducted verification tests to validate engineering solutions, ensuring it met required design specifications
- Developed and tested accessories to interface with safety syringes from an initial concept to production
- Created in-depth technical customer-facing documentation to ensure effective setup of automation systems
- Implemented GD&T and exact constraint principles to add design intent and improve manufacturability
- Sourced components from suppliers that met design criteria, cost requirements, and safety standards
- Effectively communicated designs and technical details to an interdisciplinary audience during design reviews

SharkNinja, Needham, MA

Mechanical Engineering Co-op, Ninja Heated New Product Development Team — January 2023 – June 2023

- Designed CAD models to enhance user experience such as by reducing touchpoint temperature by 50°C
- Employed mixed use of machine shop, power, and hand tools to fabricate parts to test theoretical solutions
- Disassembled and modified units to baseline product iterations, verify testing and cooking performance
- Conducted thermal maps, air speed, and normal temperature tests to measure product performance
- Communicated with overseas and cross-functional teams to efficiently disperse workload and meet deadlines

Projects

Critter Catcher, Senior Capstone Project — July 2024 – December 2024

- Led mechanical design of a 100+ part vacuum-based insect capture device in SolidWorks through 6 design iterations, managing sub-system integration and hardware sourcing using SolidWorks PDM
- Conducted a 3-stage ergonomic study with 20+ participants to optimize handle geometry and grip material
- Validated prototype device performance through field testing with live insects at a local apiary
- Earned “Best Iterative Design” award out of 26 teams by leading mechanical design decision-making through rapid prototype iteration and cross-functional team coordination

Dual Rotary Encoder Macropad, Personal Project — March 2026

- Designed and 3D-printed a custom dual-encoder macropad with a magnetic quick-release monitor mount
- Programmed Arduino firmware using HID library enabling dual-mode use for volume, media, and arrow keys
- Used heat-set inserts, locating pins, and color-coded encoder rings for easy assembly and knob identification

Miniature Product Recreations, Personal Project — April 2023 – Ongoing

- Designed and fabricated miniature models of co-op projects using FDM printing and multi-part assembly
- Designed and fabricated 10 miniatures ranging from simple single-part models to 20+ part assemblies